

3	(a)	Work done is the product of the force and the displacement moved in the direction of the force	B1
		Marker's comment Many students left out the "in the direction of the force".	
	(b)	(i) Displacement = Area of $v - t$ graph $= \frac{1}{2} (1.0 + 4.0) \times 2.4$ $= 6.0 \text{ m}$	M1 A1
		Marker's comment Many students left the answer in 1 sf. Students should know that the readings should be read to half of the smallest division, and hence to 1 d.p. Therefore, the displacement should be calculated to 2 s.f. BOD is given. The marking of this question is quite lenient. Students who showed workings to find area under the graph, 1 mark is given, although the range could be incorrect.	
		(ii) Increase in GPE = $mg\Delta h$ $= 13000 \times 6.0$ $= 7.8 \times 10^4 \text{ J}$ Increase in KE = $\frac{1}{2}m\Delta v^2$ $= \frac{1}{2} \times \frac{13000}{9.81} \times (2.4^2 - 0^2)$ $= 3817 \text{ J}$ Work done against friction = $F \times s$ $= 2000 \times 6.0$ $= 1.2 \times 10^4$ Work done by motor = Increase in energy of the lift + Work done against friction $= \Delta \text{GPE} + \Delta \text{KE} + W$ $= 78000 + 3817 + 12000$ $= 9.38 \times 10^4 \approx 9.4 \times 10^4 \text{ J}$	C1 C1 C1 A1
		Marker's comment Very poorly done. Many students attempted to use the approach of determining the work done by the forces instead of energy approach. However, many considered work done by the net force rather than work done by the tension. For those who considered work done by tension, many did not realise that the tension from $t=0$ to $t=3$ is different from $t=3$ to $t=4$.	

4	(a)	(i)	Resultant force acts perpendicular to the velocity of the object	P1
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